

EXPERIMENT – 18

GENERATION OF PWM USING TIMER INTERRUPT

AIM: To generate PWM using Timer interrupt.

OBJECTIVES:

1. To configure Timer0 for periodic interrupts.
2. To generate PWM on pin P1.0 using the timer interrupt.
3. To obtain different PWM duty cycles by changing the duty value.

PROGRAM:

```
#include <reg51.h>
sbit PWM = P1^0;
unsigned char count = 0;
unsigned char duty = 100;
void timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC;
    TL0 = 0x66;
    count++;
    if(count < duty)
        PWM = 1;
    else
        PWM = 0;
    if(count >= 200)
        count = 0;
}
void main()
{
    TMOD = 0x01;
    TH0 = 0xFC;
    TL0 = 0x66;
    IE = 0x82;
    TR0 = 1;
    while(1);
}
```

OUTPUT IMAGE:



RESULT:

Thus the PWM was generated using Timer Interrupt for various duty cycle.